

# You B Tech

## premium personal portfolio website

Product Requirements Document

Generated: 2 August 2026

### Brand Color Palette Spec:

Primary: #3B82F6 | Secondary: #64748B | Accent: #06B6D4  
Background: #0F172A | Surface: #1E293B | Text: #F8FAFC

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## 01: Product Overview

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## **\*\*Problem Statement\*\***

Creative professionals (developers, designers, digital artists) need a portfolio that transcends static grids and instead immerses visitors in a narrative driven, cinematic environment.

### **\*\*Solution Summary\*\***

A Next.js based personal portfolio website that presents a living pixel art world as a full screen animated backdrop, with a lightweight, elegant UI floating above. The experience emphasizes storytelling, dark atmospheric aesthetics, and seamless micro interactions while remaining performant and accessible.

**\*\*Target Audience\*\***

- Independent developers, designers, and digital creators seeking to showcase work in an engaging manner.
- Employers and recruiters looking for polished, memorable candidate presentations.
- Fans of indie game style visual storytelling who appreciate retro futuristic design.

## **\*\*Core Value Proposition\*\***

- Immersive cinematic pixel art world that frames all content.
- Dark, atmospheric palette with high contrast for readability.
- Story first layout that guides users through projects, skills, and contact actions.
- Full screen responsive design with dark mode toggle, SEO optimized routing, and WCAG AA accessibility.

### \*\*Measurable Success Goals\*\*

1. Achieve a Lighthouse Performance score "e /90 on mobile.
2. Maintain an average session duration "e /2 /minutes (!' /30 /% vs. baseline).
3. Reduce bounce rate to "d /20 /% through engaging micro interactions.
4. Generate "e /15 /% conversion from visitors to contact form submissions.
5. Rank within the top 3 search results for "personal portfolio developer" within 6 /months.

## 02: Features & Requirements

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## \*\*Prioritized Functional Requirements\*\*

1. **\*\*Hero Section\*\***
  - Full screen looping pixel art animation.
  - Prominent name, tagline, and primary CTA (e.g., "Explore My Work").
2. **\*\*Navigation Bar\*\***
  - Sticky, dark mode toggle, links to all core sections.
  - Responsive hamburger menu for mobile.
3. **\*\*About Section\*\***
  - Narrative bio with optional avatar illustration.
  - Scroll triggered reveal animations.
4. **\*\*Featured Projects Grid\*\***
  - Card based showcase with hover/focus micro interactions.
  - Each project links to a detailed view (title, description, media, tech stack, live demo link).
5. **\*\*Skills Overview\*\***
  - Icon based skill categories with animated progress indicators.
6. **\*\*Experience Timeline\*\***
  - Chronological visual timeline of employment and freelance work.
7. **\*\*Services Section\*\***
  - Brief bullet list of offered services with brief descriptions.
8. **\*\*Testimonials Slider\*\***
  - Rotating client or peer endorsements with quotation styling.
9. **\*\*Contact Form\*\***
  - Name, email, message fields; simple submission handling via serverless function or third party endpoint.
  - Real time validation and success/failure feedback.
10. **\*\*Footer\*\***
  - Copyright, social links, minimal navigation, and optional newsletter signup.
11. **\*\*Dark Mode Toggle\*\***
  - Persistent user preference stored in local storage.
12. **\*\*Accessibility Enhancements\*\***
  - Keyboard navigable components, ARIA labels, focus outlines.
13. **\*\*SEO Foundations\*\***
  - Server side rendering, meta tags, structured data (JSON LD) for person and software application.

**\*\*Priority Levels\*\***

- Must have: Hero, Navigation, About, Projects Grid, Contact Form, Dark Mode, SEO basics.
- Should have: Skills, Experience Timeline, Services, Testimonials, Responsive breakpoints.
- Could have: Advanced animation effects, optional newsletter, experimental UI experiments.

[illegible]

## Brand Color Palette & Styling Tokens

| Token       | Hex Code      |
|-------------|---------------|
| Primary     | #3B82F6       |
| Secondary   | #64748B       |
| Accent      | #06B6D4       |
| Background  | #0F172A       |
| Surface     | #1E293B       |
| Text        | #F8FACD       |
| Font Family | Pixelify Sans |

## **\*\*Design Direction\*\***

- Interface components (buttons, cards, inputs) use the **Primary** color for primary actions, **Accent** for highlights, **Secondary** for subdued UI elements, and **Surface** for background layers.
- **Text** uses the **Text** token for high readability on dark surfaces.
- The **Background** token sets the overall page backdrop; **Surface** is used for secondary panels and modals.
- All interactive elements respect a 1.5 /px focus ring using **Accent** to ensure keyboard visibility.

## **\*\*Core Screens\*\***

- Hero (full screen animated background)
- Navigation Bar
- About Section
- Projects Grid & Detail Pages
- Skills Overview
- Experience Timeline
- Services Section
- Testimonials Slider
- Contact Form
- Footer

## \*\*Responsive Breakpoints\*\*

- **\*\*Mobile\*\*:** "d /640 /px – stacked layout, simplified navigation, larger tap targets.
- **\*\*Tablet\*\*:** 641 1024 /px – two column grid for projects, adjusted font sizes.
- **\*\*Desktop\*\*:** > /1024 /px – multi column layouts, richer animations, expanded content.

## \*\*Accessibility Expectations\*\*

- Contrast ratio "e /4.5:1 for normal text, "e /3:1 for large text against **\*\*Surface\*\*** and **\*\*Background\*\***.
- All interactive components reachable via keyboard; visible focus state using **\*\*Accent\*\***.
- Images and animations include descriptive `alt` text or aria descriptions.
- Content is structured with semantic HTML (`<header>`, `<nav>`, `<main>`, `<section>`, `<footer>`).
- Language declared (`<html lang="en">`) and reading direction set to left to right.

## \*\*Micro Interaction Guidance\*\*

- Use ``framer-motion`` for smooth entrance/exit animations.
- Hover states should transition using the **Primary** and **Accent** colors with a 0.2 /s ease out curve.
- Scroll triggered reveals should respect reduced motion preferences.

## 04: Technical Requirements

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### **\*\*Technology Stack\*\***

- Frontend: Next.js 14 (App Router)
- Dark Mode Management: `next-themes`
- Animation: `framer-motion`
- SEO Enhancements: `next-sitemap`, built in `next/font` for Pixelify Sans
- Data Storage: Static JSON files in `/data` loaded at build time (no backend)

### **\*\*Recommended Libraries\*\***

- `next-themes` – persistent dark mode support.
- `framer-motion` – declarative animation for pixel art elements and UI components.
- `next-sitemap` – automatic sitemap generation for SEO.
- `@radix-ui/react-icons` – optional icon set for skills and services.

### **\*\*Folder / Architecture Overview\*\***

...

/app

/layout

RootLayout.tsx // global providers, ThemeProvider

/page

index.tsx // Hero

about.tsx

projects/

[slug].tsx // Project detail

skills.tsx

experience.tsx

services.tsx

testimonials.tsx

contact.tsx

footer.tsx

/components

Hero.tsx

Navigation.tsx

ProjectCard.tsx

SkillSet.tsx

Timeline.tsx

ContactForm.tsx

Footer.tsx

/data

content.json // aggregated content (projects, experience, skills, etc.)

public/

/assets

pixelart-bg.mp4 // looping background video or sprite sheet

fonts/

PixelifySans/\*.woff2 // self hosted Pixelify Sans files

...

### **\*\*Data Flow\*\***

- Content is defined in `/data/content.json`.
- At build time, Next.js `getStaticProps` (or `fetch` in Server Components) reads the JSON and passes it as props to components.
- No API routes are required; all data is rendered statically.
- For the contact form, a lightweight serverless function (`/api/contact`) can forward submissions to an external service (e.g., Formspree) or use `emailjs`.

### **\*\*Key API Endpoints\*\***

- `GET /api/contact` – receives POST from contact form, returns success/failure JSON.
- No other backend endpoints; all UI state managed client side.

### **\*\*Data Schema (content.json)\*\***

```

``json
{
  "hero": {
    "title": "Your Name",
    "tagline": "Creative Developer & Designer",
    "ctaLabel": "Explore My Work"
  },
  "projects": [
    {
      "id": "1",
      "title": "Project One",
      "description": "Brief description.",
      "media": ["/images/project1-1.png", "/images/project1-2.png"],
      "tech": ["React", "Next.js"],
      "demoUrl": "https://example.com/project1"
    }
    // ...additional projects
  ],
  "skills": [
    {"name": "React", "icon": "/icons/react.svg"},
    {"name": "Next.js", "icon": "/icons/next.svg"},
    {"name": "Pixel Art", "icon": "/icons/pixel.svg"}
  ],
  "experience": [
    {
      "position": "Senior Developer",
      "company": "Company X",
      "date": "2021–Present",
      "details": ["Led a team of 5", "Delivered 10+ features"]
    }
  ],
  "services": [
    {"title": "Web Development", "description": "Full stack solutions."},
    {"title": "Design Systems", "description": "Scalable component libraries."}
  ],
  "testimonials": [
    {
      "quote": "Excellent collaboration on the redesign.",
      "author": "Jane Doe, Client"
    }
  ],
  "contact": {
    "email": "you@example.com"
  }
}
``

```

#### **\*\*Performance & Build Considerations\*\***

- Optimize the pixel art background using compressed WebM/MP4 with `next/optimize/image` fallback.
- Enable `next-font` to preload Pixelify Sans with `font-display: swap`.
- Use `next-image` for all static assets to serve appropriately sized images.
- Generate a static sitemap with `next-sitemap` for search engine indexing.

#### **\*\*Accessibility Implementation\*\***

- Apply `aria-label` and `role` attributes to navigation and interactive components.
- Ensure focus outlines are visible using the **\*\*Accent\*\*** token.
- Test with Lighthouse Accessibility audit; target WCAG /AA compliance.

#### **\*\*Deployment\*\***

- Deploy to Vercel (or any static host) – no serverless backend required beyond optional contact function.
- Enable automatic HTTPS, CDN caching, and incremental static regeneration for content updates.

===FILE:04\_Technical\_Requirements===